The Journal of Academic Librarianship 50 (2024) 102942

Contents lists available at ScienceDirect

The Journal of Academic Librarianship

journal homepage: www.elsevier.com/locate/jacalib

# Comparing generative artificial intelligence tools to voice assistants using reference interactions

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ARTICLE INFO

*Keywords:*
Artificial intelligence
Voice assistants
Generative AI
Reference

ABSTRACT

This study investigates the ability of voice assistants and generative AI tools to respond to reference questions traditionally received by academic librarians. The authors created a sample of 25 questions based on queries received on the virtual reference service at their institution. They then created a rubric to evaluate the quality of the answers that the AI powered tools provided. The authors determined that the tools understand reference questions well and provide relevant answers but that the quality of the references provided, and the accuracy of the answers can be lacking. They suggest that more research needs to be done to understand the place of AI powered tools in reference services.

## Introduction

Reference has been an intrinsic part of university library services for decades and yet users sometimes turn to other means to answer their reference questions. From search engines to Wikipedia, users are not opposed to trying new technologies to quickly answer their inquiries and the introduction of voice assistants on the commercial market has provided them with a new avenue to ask their questions.

With the growing popularity of artificial generative intelligent (AGI) tools, academic institutions have been faced with many moral quandaries by students and faculty alike. While the primary area of concern is academic integrity, academics libraries are also noting the use of ChatGPT as a reference agent for the use of literature reviews and other parts of the research process (Forrestal, 2023). This research study aims to evaluate the use of voice assistants as information seeking agents during the reference process. A cross-comparison of Alexa, Google Assistant, Siri, and ChatGPT models 3.5 and 4 were completed using 25 sample questions to test the applications responses to reference questions. The following paper explores the potential impact that voice assistant tools have on existing reference models for academic librarians. This study is a progression of research presented at the 2019 XYZ Conference on voice assistants and reference capacity (author).

## Literature review

### *Voice assistants and technology capacity*

The first documented voice assistant or speech recognition device can be traced back to 1952 with the invention of Audrey by Bell Laboratories. Audrey was designed to recognize phonemes so that telephone customers could voice dial, however, this feature was cumbersome, expensive and did not catch on with the public (Li & Mills, 2019). Almost 60 years later in 2010, Apple announced the release of Siri, a software agent designed to interpret voice commands and retrieve relevant results based on its store of information or connection to a live browser (Hoy, 2018). In the following years, Microsoft, Amazon, and Google released their own versions of a voice-driven assistant, Cortana, Alexa, and Google Assistant respectively.

Voice assistants' function when they receive a command in the form of a voice recording, the command is then interpreted by a computer who sends an answer back to the voice assistant to relay to the user (Soofastaei, 2021). The most popular use cases for voice assistant devices were revealed in a study by Ammari et al. (2019) to be "(1) Music; (2) Hands-free search; and (3) Internet of Things (IoT) control (e.g., controlling smart lights using voice commands)" (p. 17:2).

These artificially intelligent (AI) applications are often overlooked and underdeveloped compared to other uses of natural language processing and machine learning in popular products, however, among Americans, 42.7 % of the population use voice assistant devices as of

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https://doi.org/10.1016/j.acalib.2024.102942
Received 30 December 2023; Received in revised form 3 May 2024; Accepted 15 August 2024
Available online 26 August 2024
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2023, with the number expected to grow over the coming years (Insider Intelligence, 2023b).

*ChatGPT and technological capacity*

This model of information seeking through AI applications was disrupted in 2022 with the release of ChatGPT, a tool built using artificial generative intelligence (AGI) by OpenAI. ChatGPT differs from voice assistants as it is not wholly a search and retrieve tool, but an application that can generate text based on the large language model (LLM) it was trained on (Wu et al., 2023). This LLM is represented by over 45 terabytes of training data made up of content from all over the known web and beyond (Abdullah et al., 2022). One drawback of ChatGPT and other generative AI tools is the concept of hallucinations. Hallucinations are instances where no answer can be found by the AI and so "rather than state that it has no answer, which it can do, the program will instead manufacture a response" (Frederick, 2023, p. 7).

Voice assistants are also trained with large models of data but are not designed to generate content based on prompts. This distinct feature of generative content has allowed ChatGPT, and other AGI tools such as BingAI and Google Bard, to skyrocket in popularity. In September 2023, Open AI announced that the subscription-fee version of ChatGPT, ChatGPT 4, would now have voice interaction capabilities, among other new features, that would allow users on the app to have back and forth voice conversations. (OpenAI, 2023).

This advancement opens the door for ChatGPT 4 to become embedded in everyday use on a smart phone device, similar to voice assistants. ChatGPT users reached 18.1 % of Americans in 2023 and are forecasted to make significant user adoption increases (Insider Intelligence, 2023a).

*Popularity of voice driven features*

The importance of speech plays a fundamental role in voice assistant devices. Nass and Brave (2005) wrote "Speech is the fundamental means of human communication. Even when other forms of communication – such as writing, facial expressions, or sign language – would be equally expressive, (hearing people) in all cultures persuade, inform and build relationships primarily through speech" (p.1). This is echoed years later by Phan (2017), who states "Spoken language falls under the larger umbrella term of natural user interfaces, a term that takes as its dialectic definition of the 'natural' a sense of the innate and already present functions of human action" (p. 28). Phan's assertions are in direct relation to the use of voice in virtual language and suggest that voice-drive interfaces provide an ease of integration unlike other devices because of the naturalness of speech.

Similarly, Humphry and Chesher (2021) suggest that the invocationary acts, which are a proposed structure of speech acts (Austin, 1975) performed as part of the interactions with voice assistants, further reinforces the natural within these devices. By participating in invocationary acts, humans are reminded of other such conversations and therefore deem the interaction to be natural. This naturalness can be further reinforced by the humanoid or gendered voices used by voice assistants. However, the use of female voices has been the subject of criticism for many years (Bergen, 2016; Cambre & Kulkarni, 2019; Dawnson, 2023). Though recent years have seen developments from the company creators to provide more gender variation in voice options.

Other voice features of smart phones are also quickly becoming popular such as voice messaging or voice notes. These features allow users to use speech-to-text features or send audio recordings and have been on the rise over the past few years with the application WhatsApp declaring that over 7 billion voice messages are sent daily on the platform. (WhatsApp, 2022). Further, a study by Kumar and Epley (2021) on the impact of communication media among people found that voice-based media interactions resulted in stronger connections among participants. It then comes as no surprise that OpenAI would want to capitalize on voice features for their own conversational application.

*Reference and chatbots*

In 1986, Smith posed the question "If a computer could diagnose bacterial infections and prescribe treatment why couldn't a computer diagnose an information need and prescribe a reference book to solve the problem?" (p. 487). Nearly four decades ago, librarians were considering how intelligent agents might impact reference services. Smith's work reveals some of the major problems with building intelligent agent systems for libraries and lingers on the idea that instruction would still be needed to teach the use of these tools.

Intelligent agents, or chatbots as we know them now, are computer applications that converse textually back and forth with a user. In a study by Allison (2012), the use of chatbots to replace the reference interview was explored in an early attempt to implement chatbots by the University of Nebraska-Lincoln. Pixel, the chatbot, was a locally hosted SQL database that provided interactive sessions with users to answer questions about library services (Allison, 2012). More recent examples of in-house chatbots include San Jose University, who built Kingbot using Google Dialogflow (Rodriguez & Mune, 2022).

While many of the chatbots have been built in-house by application development teams in libraries, Springshare, the library web services provider, recently released a chatbot application that can be applied to virtual reference services (Springshare, 2023). Penn State University ran a pilot from May to August 2023 and found that only 10.1 % of chatbot interactions requested follow-up with a librarian (Reinsfelder & O'Hara-Krebs, 2023).

The impact of ChatGPT as a reference agent is already starting to be explored, with most predictions falling toward the use of the tool for personalized recommendations and 24/7 reference support (Aithal & Aithal, 2023; Chen, 2023; Panda & Kaur, 2023). Zayed University in the United Arab Emirates has already begun experimenting with a chatbot building using ChatGPT's API. This bot, Aisha, confirms the predictions of the use of ChatGPT for reference as they state "The chatbot has tremendous potential to speed up reference services and make them more accessible, providing library users with round-the-clock assistance" (Lappalainen & Narayanan, 2023, p.50).

## Research questions

This study investigates whether voice assistants and generative AI tools can successfully provide reference services. In order to determine this, the authors will user the following two questions to help frame their study:

1. Can voice assistants and generative AI answer academic library reference questions?
2. Which AI tool, Siri, Alexa, Google Assistant, or Chat GPT, provides the best answers to reference questions?

## Methodology

To answer these research questions, the authors created 25 sample reference questions inspired by queries received through the virtual reference service of their university library. The sample questions are listed in Table 1. They asked the voice assistant tools and the two models of ChatGPT each question and recorded the answers in a spreadsheet. The researchers used an Amazon Echo Dot smart speaker, a Google Home smart speaker, and an iPhone smart phone to access Alexa, Google Assistant, and Siri respectively. While voice assistants provide audio responses, it's also possible to find a transcription of the answers by using the smart phone applications of the tools. The authors were able to record the responses by taking screen captures of the voice assistant applications which they later transcribed in the spreadsheet. ChatGPT 3.5 is an online tool that responds to textual prompts and provides

**Table 1**
Sample reference questions.

| ID | Question |
|---|---|
| Q1 | How do I find industry reports on wearable technology? |
| Q2 | What are some recent statistics on refugees in Quebec, Canada? |
| Q3 | How do I cite a peer-reviewed article in APA? |
| Q4 | How do I cite a book with multiple authors in Vancouver Style? |
| Q5 | Where can I watch the film Invisible Nation? |
| Q6 | How do I conduct a literature review? |
| Q7 | What is Interlibrary Loan? |
| Q8 | How did the Arab Spring start? |
| Q9 | What is the impact of climate change on policy? |
| Q10 | How do I find information about volcanoes in Iceland? |
| Q11 | How do I find commentaries on Psalm 23? |
| Q12 | How are women portrayed in the bible? |
| Q13 | What are the top academic journals for finance? |
| Q14 | How much money did Nike make in 2020? |
| Q15 | How does the government of Cuba work? |
| Q16 | Where can I find Aristotle's On Rhetoric written in ancient Greek? |
| Q17 | Is there evidence that Shakespeare was a woman? |
| Q18 | When did the Library of Alexandria burn down? |
| Q19 | Why did the Roman Empire collapse? |
| Q20 | Where do I find maps of Toronto in 1870? |
| Q21 | What is quantum entanglement? |
| Q22 | What is the difference between a scoping review and a systematic review? |
| Q23 | How do I connect Microsoft Word to Zotero? |
| Q24 | How do I export references in EndNote? |
| Q25 | Which databases are the best to search for social science subjects? |

written responses that the researchers then copied and pasted in the evaluation spreadsheet. When using the smart phone application, ChatGPT 4 can operate with a voice prompt that is then turned to text with editing capabilities in case it was misheard. The user then sends the prompt when it is done. An alternative version of this is a voice only interaction. In the case of this project, no editing was done after a prompt was read, all questions were understood by the application and the researchers recorded the textual answers in their evaluation spreadsheet.

The authors conducted this study over one year from November 2022 to November 2023. They first tested the reference questions with the voice assistants in November 2022 and recorded the answers. With the rise in popularity of ChatGPT 3.5, they then decided to include this tool in the research project and tested it with the same sample reference questions in January 2023. In November 2023, the researchers purchased a subscription to ChatGPT 4 to test its capacity to respond to reference questions and compare the results with those of ChatGPT 3.5.

The authors created a grading rubric to measure the quality of the responses produced by AI tools. The rubric was used to assess if the tool understood the questions asked and if a complete answer that is accurate, relevant, and used authoritative sources was produced. Each category was assigned a numerical value depending on the quality of the response ranging from −1 point to 2 points. The complete grading rubric can be seen in Table 2.

Each author evaluated the responses provided by two tools, for example Alexa and Siri, meaning that each author evaluated 50 responses. The responses from ChatGPT4 were split between the two researchers. To ensure intercoder reliability, the authors did a calibration test where they coded the same interactions to make sure that they were applying the rubric similarly. They also discussed any specific issues that came out of the coding to resolve any discrepancies, for example, if a response proved different from others and made it difficult to evaluate. Once the authors completed the coding, they calculated the end results to determine which AI tool had performed better in the provision of reference services.

**Table 2**
Evaluation rubric.

| Indicator | Question | Scoring/Scale |
|---|---|---|
| Accuracy | Is the answer accurate to the question asked? | 0- not accurate<br>1- somewhat accurate<br>2- accurate |
| Relevance | Is the answer relevant to the users search needs? | 0- not relevant<br>1 - somewhat relevant<br>2 - relevant |
| Authority | Is the source one of authoritative nature on the topic? | 0- not authoritative<br>1- somewhat authoritative<br>2 - authoritative |
| Recommended source | Is this a source that librarians would recommend for users? Are there any biases in this source users need to be aware of? | 0- no<br>1 - yes, with advice or caution<br>2 - yes |
| Successful comprehension | Did the VA comprehend the question asked? | 0 - no<br>1- somewhat<br>2- yes |
| Incorrect content | Was incorrect content provided in the answer? | −1 - yes<br>0 - no |
| Missing content | Were there parts of the question asked that went unanswered? | −1 - yes<br>0 - no |
| Presence of references | Are references included? | 0- no<br>1 - yes |

## Results

### *Successful comprehension*

Each question was first evaluated to determine if the reference question asked could be comprehended by the device. Many instances of devices failing to understand voice prompts or regional accents have been recorded and so it was important to note if the device could comprehend the question being asked. The results of device comprehension are explored in Fig. 1.

ChatGPT 4 had 96 % success in comprehending the question in its entirety. Siri was unsuccessful in fully comprehending two of the questions (Q16 and Q20) and only partially understood one question (Q23). All other questions were successful.

### *Incorrect content*

Overall, all tools provided correct content and scored above 84 % in this category. Google Assistant and ChatGPT 4 provided the most correct answers with 96 % and Siri provided 88 % correct content. Alexa and ChatGPT 3.5 had a slightly larger margin of error with 84 % of correct answers.

### *Accuracy*

For each of the 25 reference questions, the researchers analyzed them to determine if the responses provided were accurate to the question's topic. Fig. 2 shows the grading evaluation of each device tested and its percentage of accuracy. Google Assistant was the top performer with an 88 % accuracy rating and 12 % somewhat accurate, no responses were deemed inaccurate for this device. ChatGPT 3.5 was the lowest performer with 16 % inaccurate responses.

### *Relevancy*

The relevancy of the answers provided by the devices to the reference questions was evaluated and is represented in Fig. 3. ChatGPT 4 had a 96 % average of relevant responses to reference questions, closely followed by Google Assistant with 92 %. Siri had the least relevant

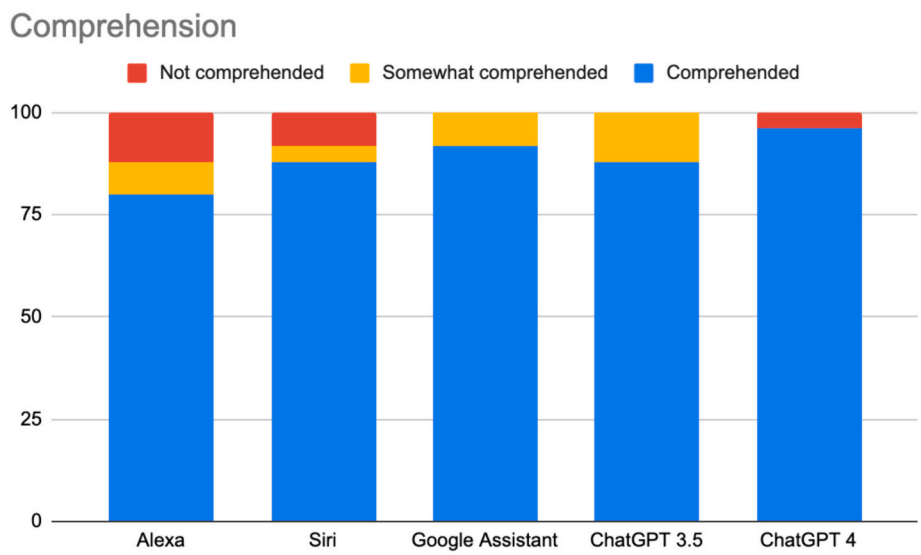


**Fig. 1.** Voice assistant and ChatGPT model ability to comprehend the questions asked vocally by researchers.

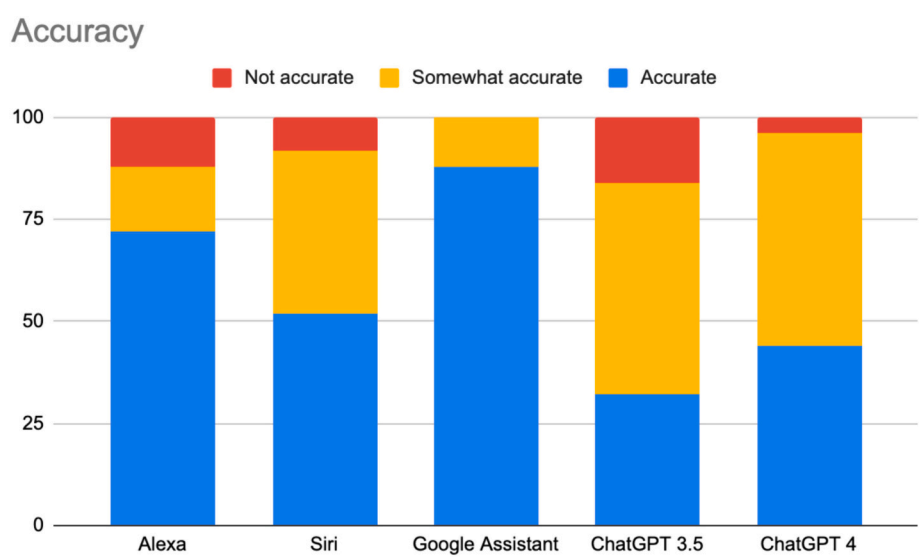


**Fig. 2.** Voice assistant and ChatGPT model accuracy in responding to reference questions.

responses at 76 %.

*Authority*

Each source provided by the devices was evaluated to determine the authority of the resource related to the reference question asked. The results of authoritative sources are reflected in Fig. 4. Siri and the Google Assistant were tired with a 40 % average for authoritative responses to questions, however, with the inclusion of a 56 % average for somewhat authoritative, Google Assistant is the top performer. No responses from ChatGPT 3.5 were considered authoritative as this model does not include citations.

*Recommended source*

Of sources provided, each was reviewed to determine if an academic librarian would recommend the provided resources as a response to the reference question asked. The results of the recommended sources at explored in Fig. 5. Google Assistant was again the top performer with 40 % recommended sources and 56 % recommended with caution. Only one Google Assistant response (Q19) had no sources that would be recommended in some capacity by a librarian. Although ChatGPT 3.5 was not considered an authoritative source since it did not provide references, the authors judged that the information it provided could be used in some instances. For example, when ChatGPT 3.5 provided definitions for library services or advice on how to use a specific citation style, the authors determined that the user could rely on the information if they confirmed it by checking another source.

*Missing content*

None of the tools included high rates of missing information. Google Assistant, ChatGPT 4, and Alexa provided all the content needed in 96 % of questions. Siri and ChatGPT 3.5 omitted more content and responded completely to 88 % of questions.

*Presence of references*

A big disparity can be seen in the presence of references with some tools providing references of excellent quality and others omitting

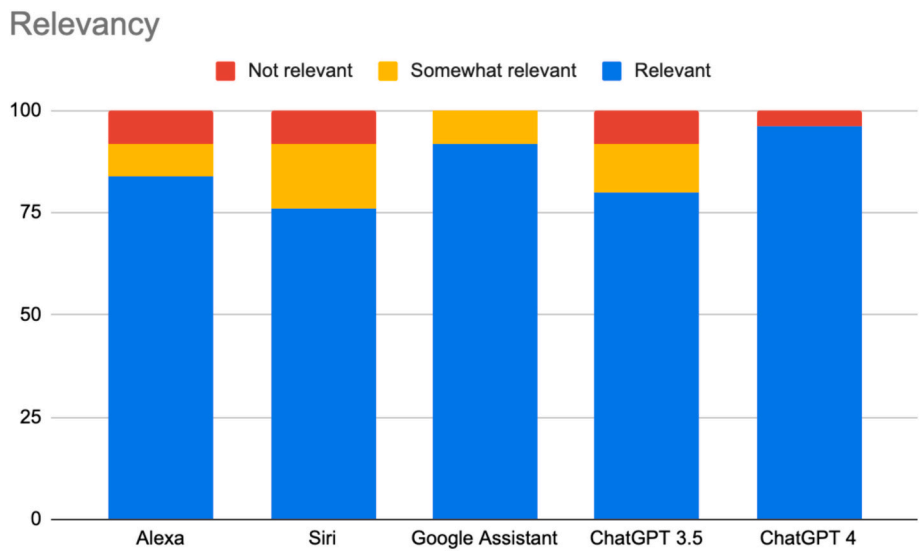


**Fig. 3.** Voice assistant and ChatGPT model ability to provide relevant answers to reference questions.

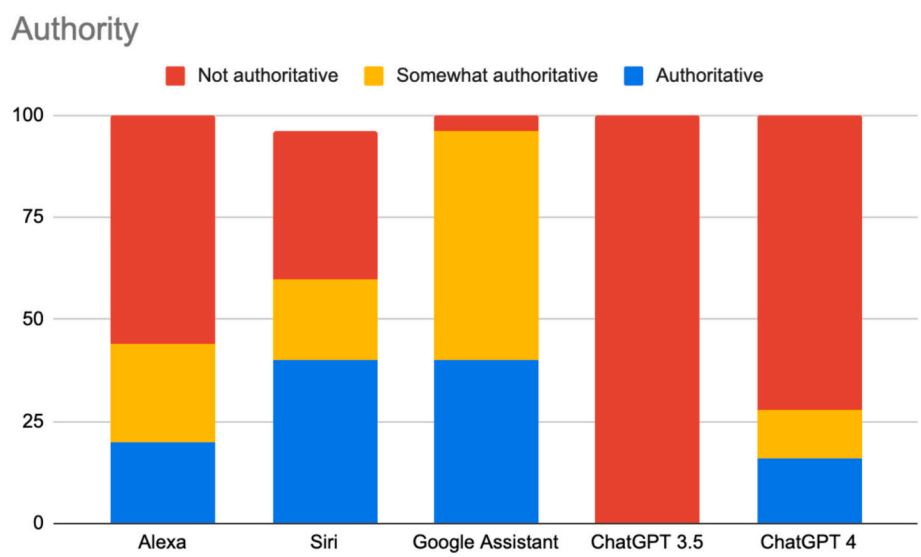


**Fig. 4.** Voice assistant and ChatGPT model ability to provide authoritative sources to in response to reference questions.

references completely. Siri and Google Assistant performed very well by including references of good quality in 96 % and 92 % of instances respectively. Alexa came in third position with 76 %. A big drop can be seen with both instances of ChatGPT; the 3.5 model including no references in all interactions and model 4 including references of reputable quality for 36 % of the questions.

## Discussion

*Can voice assistants and generative AI answer library reference questions?*

Overall, these AI tools are proficient at understanding user questions and providing relevant answers for them. The natural language processing capabilities of each tool are sophisticated enough that they can understand the user's information need accurately and provide relevant information to answer this need. However, the accuracy and authority of the answers vary greatly. The greatest disparity in performance can be seen in the accuracy and authority of the responses, as well as whether they would be recommended by a librarian as a resource.

Another concern that emerges from the results is that some of the tools do not provide references for their answers or provide references of poor quality. Voice assistants did better by a wide margin for references, providing better results than the two iterations of ChatGPT. The absence of references in ChatGPT is a well-known cause for concern among information professionals and these results confirm the presence of hallucinations, fake citations or a lack of citations in answers to reference questions (Forrestal, 2023).

While social scientists argue that speech holds an important place in our lives and that it is reflected in our use of technology, it seems that the voice assistants are not only pleasant to use but provide the best answers to library inquiries in terms of quality (Phan, 2017). Google Assistant, Siri, and Alexa provided the best answers in terms of relevance, accuracy, and the presence of references. Since voice assistants retrieve their answers from internet searches, and do not depend on a large language model, they are less likely to provide fake citations (Soofastaei, 2021). These findings are in keeping with Ammari et al. (2019) who determined that hands-free searching is the second most popular use of voice assistants.

Even though these tools can provide relevant and accurate answers to reference questions, they would not provide answers that are

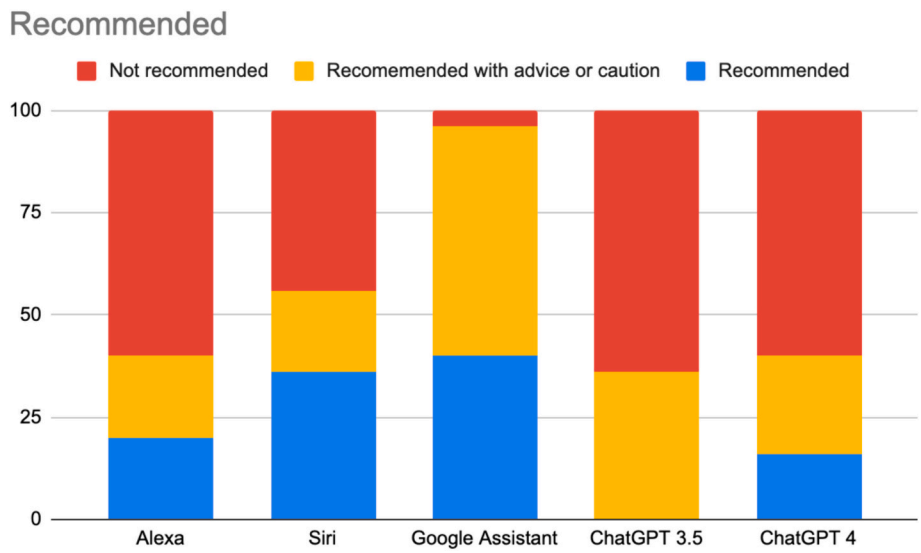


**Fig. 5.** Sources recommended by librarians as provided by voice assistant devices and ChatGPT models.

comparable to what an academic librarian can offer. Both instances of ChatGPT and all the voice assistants struggled with whether the sources would be recommended by an academic librarian. Google Assistant's answers would be recommended in only 40 % of instances with the other tools providing recommended answers in even fewer instances. Over 50 % of the sources provided by both instances of ChatGPT would not be recommended by an academic librarian. While the other voice assistants had lower percentages of "not recommended", Google Assistant is the only one that showed a high percentage of "recommended with advice or caution", which means that an academic librarian would warn the user to properly evaluate the source depending on its authority in the particular context they want to use it in.

*Which AI tool, Siri, Alexa, Google Assistant, or Chat GPT, provides the best answers to reference questions?*

Overall, Google Assistant showed the best performance across all indicators. It also provided the best references of all the tools. It is interesting to note that it often referred to academic library guides on various subjects such as citations thus making it more authoritative than the other tools that selected webpages of lesser quality. This may be influenced by user profiles that frequent academic library guides; however, it shows the power of search results when trained by an information literate user. While Alexa understood most of the questions and provided somewhat accurate responses, it often selected poor sources. In comparison, Siri provided less accurate responses but performed better in terms of the authority of the sources provided and whether they would be recommended by an academic librarian.

These findings regarding voice assistants confirm the results of a previous study conducted by the authors (author). Both instances of ChatGPT performed well in terms of comprehension of the questions asked and relevancy of the responses but the accuracy and quality of the sources was too weak to make them a contender in terms of providing reference services. Lai (2023), using different research methods, also found that ChatGPT cannot accurately provide answers to reference questions.

For the most part all the tools provided correct information and little information was missing but as previously mentioned, the quality of the information and the sources cited varied greatly. It is also important to note that much like many technological tools, voice assistants and ChatGPT frequently undergo updates to their coding which would hopefully increase the quality of the answers that they provide. This study aims to deliver a snapshot of the quality of the answers at a certain moment in time.

## Limitations

Voice assistants and ChatGPT require users to create an account to use them and as such, they collect data on the user's questions in order to provide more targeted answers and information. Given that the devices tailor their content to their users, so that they can have the best experience possible, this means that two different users with different profiles based on previous questions they have asked, may receive different answers to the same question. While academic librarians are trained to provide reference services, it's highly unlikely that two different librarians would respond to the same question in exactly the same way and as such, the authors determined that the limitations caused by the voice assistant profiles would have a minimal impact on this study and they chose to focus instead on the quality of the answers provided.

Manual coding does present some limitations as it is an inherently subjective process. In order to mitigate this subjectivity, the authors created the codebook together, ran a calibration test to ensure that they are coding similarly, and discussed any outliers together. Several previous studies have relied on manual coding to evaluate reference interactions and have produced reliable results (Armann-Keown et al., 2015; Dempsey, 2017; Hervieux & Tummon, 2018; Matteson et al., 2011).

There could be some limitations with the potential reproducibility of this study as ChatGPT 4 requires a paid subscription and the voice assistants require an initial purchase of the hardware for the tools (Alexa device, Google Assistant, etc.). Given that most voice assistants can be used without the physical tools by downloading their respective applications and that the ChatGPT 4 subscription is not prohibitively expensive (20 US$/a month), they believe these limitations to be acceptable as they would not prevent the reproducibility of the study by all potential researchers. Furthermore, due to the evolving nature of AI tools, the authors are aware that reproducing identical results could be challenging given that the tools have undergone multiple updates since then. They believe this challenge does not undermine the value of the study as reference librarians are constantly learning new things and could provide different answers to the same questions at different points in time.

## Conclusion

Given the increasing presence of voice assistants and the near

constant improvements to generative AI tools like ChatGPT, librarians may express a fear that users would employ them instead of using more traditional reference services. While Google Assistant performed well in responding to reference inquiries, it is interesting to note that most of the references that it recommended were academic library guides. It could be of interest to investigate how often voice assistants rely on library resources to provide answers. Even though voice assistants can provide an interesting starting point for research questions, this study shows that they should be used in conjunction with traditional reference services. As for ChatGPT, it can understand prompts and provide relevant answers but its inability to cite authoritative sources creates issues with reliability and authority. Perhaps other uses should be explored for these tools such as brainstorming or explaining concepts. Future studies could also explore how AI tools can be used in the provision of reference services in partnership with reference librarians.

## CRediT authorship contribution statement

**Amanda Wheatley:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Sandy Hervieux:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization.

## Declaration of competing interest

There are no declarations of interest to declare for this manuscript.

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